

CIS 6270

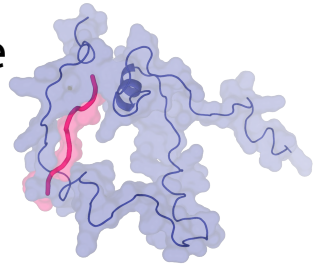
Module 1: Introduction and Math Foundations

Lecture 1.3: Maps, Pushforwards, and Changes of Variables

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Pranam Chatterjee, PhD
Assistant Professor of Bioengineering
Assistant Professor of Computer and Information Science
University of Pennsylvania

Penn Engineering
University of Pennsylvania



For a set A contained in the input space, it has an image

$$\phi(A) = \{\phi(x) \mid x \in A\}$$

The image ϕ(A) is the set of every output ϕ(x) produced by inputs x belonging to the set A.

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The inverse image $\phi^{-1}(B)$ [illegible] ϕ(x) belongs to the output set B.

An inverse image is defined even when the map has no inverse function. It just collects every input that reaches the chosen output region.

A little example

Let $$\phi(x) = 2x + 1$$

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and choose the output interval

$$B = [1.4, 2.0]$$

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Therefore,

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- One-to-one means distinct inputs have distinct outputs.
- Onto means every target value is reached by at least one input.
- A map satisfying both properties is bijective.

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Therefore,

$$\boxed{\phi^{-1}(y) = \frac{y - 3}{2}}$$

The inverse map ϕ^{-1} sends the output y back to the input by subtracting three and dividing by two.

As an example with numbers:

Let $$y = 2x + 3$$

So $$x = \frac{y - 3}{2}$$

Therefore, $$\boxed{\phi^{-1}(y) = \frac{y - 3}{2}}$$

$\phi(4) = 11$

$\phi^{-1}(11) = 4$

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There are maps that aren’t globally invertible

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If the domain is restricted to nonnegative numbers, the inverse becomes

$$\phi^{-1}(y) = \sqrt{y}$$

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Restricting the allowed inputs can turn a many-to-one map into a one-to-one map.

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What about the inverse of an affine map

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If A is invertible, subtract b:

$$y - b = Ax$$

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Then multiply by A^{-1}:

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If A is invertible, subtract b:

$$y - b = Ax$$

Then multiply by A^{-1}:

$$\boxed{x = A^{-1}(y - b)}$$

The input x is recovered from output y by subtracting the translation vector b and then multiplying by the inverse matrix A^{-1}.

A short example!

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Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

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The forward map gives

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The inverse recovers x:

$$\begin{aligned} x &= A^{-1}(y - b) \\ &= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \left(\begin{bmatrix} 5 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \end{aligned}$$

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One affine map can describe a single transformation.

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Richer transformations arise by applying several maps in sequence

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The composed map ψ o ϕ first applies ϕ to the input x to produce the intermediate value y, and then applies ψ to y to produce the final output z.

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$$(\psi \circ \phi)(4) = 11$$

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The order matters:

$$(\phi \circ \psi)(x) = 2(x + 3) = 2x + 6$$

Composition explains how several transformations move one point.

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The order matters:

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If the original input is random, applying those transformations also changes its probability distribution.

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Maps of random variables

Let X be a random variable with distribution p_X. Define:

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Applying ϕ to every sample moves the complete population of points. The distribution formed by the outputs is called the pushforward distribution.

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The probability that the output random variable Y lies in an output region B equals the probability that the input random variable X lies in the inverse image $\phi^{-1}(B)$, which contains every input mapped into B.

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To calculate the probability of an output region, trace the region backward through the map and measure the probability of all inputs that arrive there.

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The simplest pushforward calculation occurs in a finite discrete state space, where probability masses reaching the same output are <u>added</u>.

Discrete pushforward example

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$$\mathcal{X} = \{a, b, c\}$$

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Then

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Then

$$p_Y(u) = p_X(a) = 0.2$$

while

$$\begin{aligned} p_Y(v) &= p_X(b) + p_X(c) \\ &= 0.5 + 0.3 \\ &= 0.8. \end{aligned}$$

The states b and c reach the same output v, so their probability masses combine.

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The same principle applies in continuous spaces.

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Because X is uniform on [0,1],

$$\mathbb{P}(X \in [0.2, 0.5]) = 0.3$$

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Therefore

$$\mathbb{P}(Y \in [1.4, 2.0]) = 0.3$$

The input interval has width 0.3. Its output image has width 0.6. The probability remains 0.6, so the output density must be half as large.

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We have now followed probability through one map.

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Deep transformations contain several maps.

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Equivalently,

$$\boxed{(\psi \circ \phi)_{\#} p_X = \psi_{\#}(\phi_{\#} p_X)}$$

Pushforwards follow the same composition order as the point transformations.

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$$Y = \phi(X)$$

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The pushforward of p_X through the composed map ψ o ϕ equals the distribution obtained by first pushing p_X through ϕ and then pushing that intermediate distribution through ψ.

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This matters because a complicated generative transformation can be built from **many simpler transformations** while preserving a strong description of *how* probability moves through the sequence.

Composition describes a finite sequence of transformations.

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Continuous generative models extend this idea to a family of maps indexed by time.

Time-dependent maps

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Training adjusts θ so that p_1 approaches the data distribution.

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A nonlinear map can stretch, compress, rotate, or shear different regions by different amounts.

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To calculate the transformed density, we need a local description of the map near each point.

That local description begins with the derivative.

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For a scalar function f, the derivative at x is

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For a small input displacement Δx, the new output f(x+Δx) is approximately the original output f(x) plus the local slope f′(x) multiplied by Δx.

A super simple example

$$f(x) = x^2$$

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For a small displacement 0.1:

$$f(2.01) \approx f(2) + f'(2)(0.01) = 4 + 4(0.01) = 4.04$$

The local linear approximation is highly accurate for a small displacement.

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A scalar derivative handles one input and one output.

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A multivariable map has several inputs, so we need to measure sensitivity with respect to each input coordinate separately.

Partial derivatives

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$$\frac{\partial f}{\partial x_1} = 2(1) + 3(2) = 8$$

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Near this point, changing x_1 by a small amount has a larger effect on f than applying the same change to x_2.

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A vector-valued map has several outputs as well as several inputs.

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We therefore need one partial derivative for every output-input pair.

The Jacobian

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Let

$$\phi : \mathbb{R}^d \to \mathbb{R}^m$$

have output components

$$\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_m(x) \end{bmatrix}$$

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For every output ϕ_i and every input x_j, calculate:

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Arrange these partial derivatives…

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Into the Jacobian Matrix!

$$J_\phi(x) = \begin{bmatrix} \frac{\partial \phi_1}{\partial x_1} & \frac{\partial \phi_1}{\partial x_2} & \cdots & \frac{\partial \phi_1}{\partial x_d} \\ \frac{\partial \phi_2}{\partial x_1} & \frac{\partial \phi_2}{\partial x_2} & \cdots & \frac{\partial \phi_2}{\partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \phi_m}{\partial x_1} & \frac{\partial \phi_m}{\partial x_2} & \cdots & \frac{\partial \phi_m}{\partial x_d} \end{bmatrix}$$

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The Jacobian $J_\phi(x)$ is the m x d matrix whose entry in row i and column j is the partial derivative of output component ϕ_i with respect to input coordinate x_j, evaluated at x.

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This is the multivariable replacement for the ordinary derivative.

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$$\phi(x_1, x_2) = \begin{bmatrix} x_1^2 + x_2 \\ \sin(x_1) + 3x_2 \end{bmatrix}$$

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Therefore,

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At (1,0)

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$$J_\phi(1, 0) = \begin{bmatrix} 2 & 1 \\ \cos(1) & 3 \end{bmatrix} \approx \begin{bmatrix} 2 & 1 \\ 0.540 & 3 \end{bmatrix}$$

We have now collected every local input-output sensitivity into one matrix.

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The next step is to see how this matrix approximates the complete nonlinear map near a point.

Local linearization

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The nonlinear map may curve globally. At sufficiently small scales, its behavior resembles the linear transformation given by the Jacobian.

Look at it in this example

Using the previous Jacobian at (1,0), choose

$$\Delta x = \begin{bmatrix} 0.01 \\ -0.02 \end{bmatrix}$$

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The Jacobian predicts almost no change in the first output and a decrease of approximately 0.0546 in the second output.

The Jacobian describes local directional changes.

Using the previous Jacobian at (1,0), choose

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The absolute determinant measures area scaling.

When the input and output dimensions are equal, its determinant summarizes the local change in volume.

For a two-dimensional matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

the determinant is

$$\det(A) = ad - bc$$

The determinant of the two-by-two matrix A equals the product of its diagonal entries a and d minus the product of its off-diagonal entries b and c.

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It collapses every point onto a line and cannot be inverted.

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A nonlinear map has a different Jacobian at each point.

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We now know how the map changes local volume.

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The probability mass $p_X(x)\,dx$ in a small input interval near x equals the probability mass $p_Y(y)\,dy$ in its mapped output interval near y because the map relocates probability without creating or destroying it.

The derivative relates the interval widths

$$dy = |f'(x)|\, dx$$

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A transformation stretches or compresses small intervals by the factor |f′(x)|, so the density $p_Y(y)$ must change inversely to preserve the same probability mass.

Since

$$x = f^{-1}(y)$$

We obtain

$$\boxed{p_Y(y) = p_X(f^{-1}(y))\left|\frac{df^{-1}(y)}{dy}\right|}$$

Expansion spreads the same probability across more space, so density decreases.

$$dy = |f'(x)|\,dx$$

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Compression concentrates the same probability into less space, so density increases.

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Let $$X \sim \mathrm{Uniform}(0, 1)$$

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$$\int_1^3 \frac{1}{2}\, dy = 1$$

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The same conservation argument extends to several dimensions by replacing the scalar derivative with the Jacobian determinant.

Multivariable change of variables

Let

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The Jacobian determinant measures how φ locally stretches or compresses d-dimensional volume, so $p_Y(y)$ adjusts by the inverse volume change to keep total probability the same.

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The equivalent forward form is

$$\boxed{p_Y(\phi(x)) = \frac{p_X(x)}{|\det J_\phi(x)|}}$$

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The forward Jacobian tells us how much space around x expands or contracts under φ, so $p_Y(\varphi(x))$ decreases when volume expands and increases when volume contracts.

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The unit square becomes a rectangle with area 6

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Since |det(A)| = 6, the transformation expands area by 6s, so the density decreases by 6x to $p_Y(y) = 1/6$.

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This example uses one map.

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Deep transformations compose many maps

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$$\boxed{J_{\psi\circ\phi}(x) = J_{\psi}(\phi(x))J_{\phi}(x)}$$

Jacobian chain rule

The total local stretching is the combination of the stretching from φ at x and then from ψ at the intermediate point φ(x), so their Jacobians multiply.

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The Jacobian of the composed map ψ o ϕ at x equals the Jacobian of ψ evaluated at the intermediate point ϕ(x), multiplied by the Jacobian of ϕ evaluated at x.

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The log of the Jacobian determinant of a composition equals the sum of the log of the Jacobian determinants of its component maps, each evaluated at the appropriate intermediate point.

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Because volume changes multiply across successive transformations, their determinants multiply, and taking logs turns that product into a simple sum.

This additive form is especially useful for generative models built from many invertible layers.

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Log-density form

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The change-of-variables formula can be written as

$$\boxed{\log p_Y(y) = \log p_X(x) - \log \left|\det J_\phi(x)\right|}, \qquad y = \phi(x).$$

Log-density form

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$$\boxed{\log p_Y(y) = \log p_X(x) - \log \left|\det J_\phi(x)\right|}, \qquad y = \phi(x).$$

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Taking logs turns the density correction into subtraction: log $p_Y(y)$ equals log $p_x(x)$ minus the local log-volume expansion, which makes likelihoods easy to add across many flow layers.

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- The log-determinants add across composed transformations.

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A nonzero determinant guarantees local invertibility. Global invertibility requires additional properties across the complete domain.

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If several different x values produce the same y, each one contributes probability mass to y, so $p_Y(y)$ is the sum of all of their derivative-corrected density contributions.

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This explains why invertibility makes density calculations much simpler.

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This enables exact likelihood calculation for invertible normalizing flows.

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That infinitesimal change is measured by the divergence of a velocity field.

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To understand how f changes around a point, you need to calculate one partial derivative for every input coordinate.

The gradient

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The gradient points in the direction of the steepest local increase in f. Its magnitude measures how rapidly f increases in that direction.

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At $(x_1, x_2) = (1, 2)$:

$$\nabla f(1, 2) = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

The function increases much more rapidly in the x_2 direction at this point.

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So the gradient $\nabla f(x)$ points in the direction of fastest increase, and the dot product with u picks out how quickly f changes specifically along direction u.

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Moving only along x_1 asks “how much does f change if I take a tiny step to the right?”, and here that rate is 2.

A gradient is a vector derived from a scalar field.

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Continuous generative dynamics require a vector already assigned to every point

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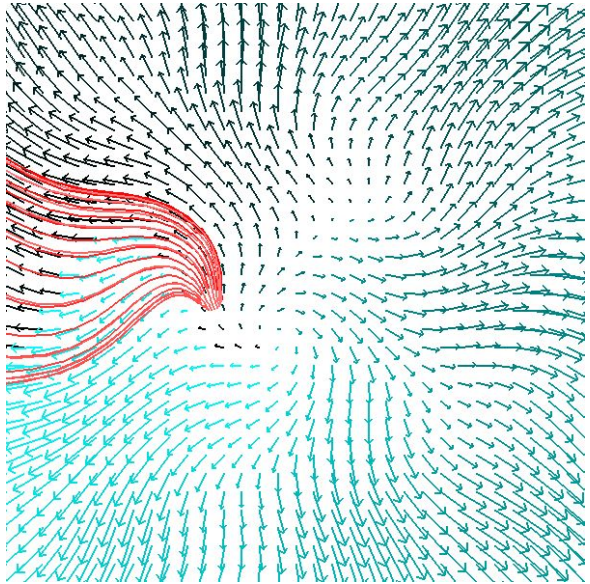
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Imagine placing a small arrow at every point in space. A sample follows the arrows, and the complete collection of samples moves through the field.

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The point moves toward the origin. Its second coordinate contracts twice as rapidly as its first coordinate.

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- Zero divergence indicates local volume preservation.

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A small region expands locally because the total divergence is positive.

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The field rotates points without locally changing area.

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The **continuity equation** connects this motion to the evolving probability density.

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Let $p_t(x)$ be the probability density and $v_t(x)$ the velocity field. Conservation of probability gives:

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It tells you how the density p_t changes for a particle that is actually moving with the flow, combining change over time with change from moving through space.

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The time derivative of the log density log p_t evaluated along a moving trajectory x_t equals the negative divergence of the velocity field v_t at that trajectory, so local expansion lowers density and local contraction raises density.

This is the continuous-time counterpart of the log Jacobian-determinant correction from change of variables.

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Score Functions

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The score $s_t(x)$ is the gradient with respect to x of the log probability density $p_t(x)$, so it points in the direction where log density increases most rapidly at time t.

For a standard Gaussian in 1D

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At x=3, the score is -3. At x=-2, the score is 2.

This quantity becomes central in diffusion because reverse dynamics require information about how noisy density changes across space.

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Diffusion also requires a *second-order* operator that measures local curvature.

Because diffusion spreads probability from high-density regions into nearby low-density regions, and that spreading depends on the local curvature of p_t, captured by the second derivative or **Laplacian** $\nabla^2 p_t$.

The Laplacian

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$$\Delta f(x) = \nabla \cdot \nabla f(x) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}$$

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- A negative Laplacian indicates that the point lies above its local surroundings.
- A zero Laplacian indicates balanced local curvature.

Laplacian example

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